

The Expected Matching Approach

A pairing system for Swiss Bracket

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Abstract

The Swiss Bracket format, widely used in competitive eSports (Counter-Strike Majors, League of legends Worlds), qualifies 8 teams from 16 through five rounds. A 2023 thread on X identified scenarios where inferior teams qualify over superior ones despite all results following expectations. We show that this failure stems from mixing incompatible proper 3-edge colorings of $K_{3,3}$ across rounds, blocking fair matching in decisive Round 5. The *Expected Matching Approach (EMA)* uses the rotation algorithm to build perfect matching on $K_{4,4}$. Across 768 000 simulated tournaments (20 samples $\times$ 8 systems $\times$ 3 solvers $\times$ 4 populations $\times$ 4 seeding classes $\times$ 100 seeds), EMA achieves 80.5% top-8 precision and 0.105 normalized utility loss, outperforming the original and current Major pairing strategies. EMA’s key advantage is Round-5 quality: It drastically improves chances of a priori fair pairings.

1 Introduction

The Swiss Bracket narrows 16 teams to 8 qualified over five rounds, used prominently in Valve’s CS:GO/CS2 Majors and Riot’s LoL World Championship as a qualifier stage. Teams with identical records are paired each round until scoring 3 wins (qualified) or 3 losses (eliminated)

In 2023, analyst Graham Pitt[1] provided an example (Figure 1) where, despite all teams performing exactly to seed, the 8th-best team fails to qualify while the 11th-best advances. Valve’s response, “speed-up” matching, could be not so much of an improvement¹.

1.1 Definition

To discuss the quality of a tournament design and matching strategies, we first introduce the notion of fairness and balance. Section 3.2 provides formal definitions, the following ones are here for the reader to understand the complaints and the problem. There are no consensual definitions; these have the benefit of being both supported by the literature[3] and accurately capturing the concerns expressed in the discussion around the example of figure 1.

¹<https://x.com/NER0cs/status/1932220367558381911?s=20>



Figure 1: Graham Pitt’s 2023 example: all teams perform exactly to seed, yet seed #8 fails to qualify while seed #11 advances.

Definition 1 (Fairness). A pairing is said to be fair if the high seeds are favorites to win their games. For example: 1 vs 16, 2 vs 15, ..., 8 vs 9.

Definition 2 (Balance). A pairing is said to be balanced if opponents in games both have similar chances to win. For example: 1 vs 2, ..., 8 vs 9, ..., 15 vs 16.

1.2 Graph Theory Background

We recall standard definitions from graph theory; see J. Bondy [5] for comprehensive treatments.

Definition 3 (Graph). A graph $G = (V, E)$ consists of a finite set of vertices V and a set of edges $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$.

Definition 4 (Complete Graph). The complete graph K_n is the graph on n vertices in which every pair of distinct vertices is connected by an edge.

Definition 5 (Complete Bipartite Graph). A graph $G = (V, E)$ is bipartite with parts A and B if $V = A \cup B$, $A \cap B = \emptyset$, and every edge connects a vertex in A to a vertex in B . The complete bipartite graph $K_{m,n}$ has $|A| = m$, $|B| = n$, and contains all mn edges between the two parts.

Definition 6 (Perfect Matching). A matching in G is a set of pairwise non-adjacent edges $M \subseteq E$ (no two edges share a vertex). A matching is perfect if every vertex of G is incident to exactly one edge in M .

Definition 7 (Proper Edge Coloring). A proper k -edge coloring of G is a map $\varphi : E \rightarrow \{1, \dots, k\}$ such that no two adjacent edges (sharing a vertex) receive the same color.

In this practical application, vertices represent teams and edges represents games. All games in a given round, or pairings, form together a perfect matching. The notion of fairness is related to perfect matching on complete bipartite graph with the set A containing the high seed teams and set B consisting of the low seed teams.

1.3 Contributions

(1) graph-theoretic root cause identification via incompatible $K_{3,3}$ edge colorings; (2) Expected Matching Approach; algorithms building proper edge with particular properties (3) Classification of proper edge coloring via cycle basis analysis (4) naming of 12 $K_{4,4}$ proper edge coloring (5) Improvements of Swiss Bracket pairing strategies validated over 768 000 simulated tournaments.

2 Problem Identification

2.1 Theory versus Reality Paradox

There exist 15 different pairings among 6 teams that correspond to perfect matching in the complete graph K_6 . They are listed by priority order in the Valve rule books². Only 6 of them are fair. They correspond to perfect match on the complete bipartite graph $K_{3,3}$. They are indeed the 6 preferred pairings by Valve. **Observation:** In figure 1, there are no fair pairings available in the last round if we do not want rematch (*). The 6 teams playing in the last round meet each other in Round 1, Round 3 and Round 5. According to König's theorem[6], a proper 3-edge coloring on $K_{3,3}$ exists. Thus, these teams can face each other fairly on three occasions, round 1, 3 and 5 (**). (*) and (**) seem paradoxical but are not strictly in contradiction. In rounds 1 & 3, the pairings involve two additional teams, seeds 7 and 10. It forms a perfect matching on $K_{4,4}$, which admit a proper 4-edge coloring, and we could play 4 fair round. Figure 2 displays the theoretical possibility alongside the state of the tournament before playing round 5 with vertices labeling:

$A = \{1, 2, 3\} = \{\text{ENCE \#5, Vitality \#6, 9INE \#8}\},$
 $B = \{4, 5, 6\} = \{\text{G2 \#9, Astralis \#11, Monte \#12}\}.$

²https://github.com/ValveSoftware/counter-strike_rules_and_regs/blob/main/major-supplemental-rulebook.md

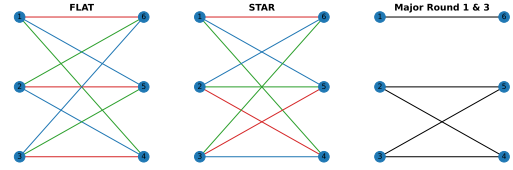


Figure 2: The two proper 3-edge colorings of $K_{3,3}$ (FLAT and STAR) and games from Round 1&3. The example pairings results in the mix of red edges from both FLAT and STAR, yet the pattern is not included in neither proper edge coloring.

2.2 Observations

These notes helped to understand and solve the apparent paradox. 2

- (1) The tournament played perfect matching from **two different proper edge coloring**, the red edges from FLAT and STAR.
- (2) **Virtual duplicate game.** An interpretation is to say that virtually, the game *1 versus 6* was played twice. In practice, there was no rematch, and games were played with two teams not present in the last round.
- (3) Two perfect matching on $K_{4,4}$, from round 1 and 3, induce a **cycle** (2,4,3,5). Edges of this cycle in FLAT and STAR are colored by three colors - not two !

These remarks lead to the construction of the following intersection (or incompatibility) graph $G = (V, E)$, $V = \{S_1, S_2, S_3, F_1, F_2, F_3\}$, with S_1 the red perfect matching of STAR, S_2 the green and S_3 the blue one (Similarly for FLAT from Figure 2) and $E = \{(u, v) | u, v \in V, u \cap v \neq \emptyset\}$ ³. This means that two perfect matches are connected if they share an edge in $K_{3,3}$. See figure 3.

2.3 Problem statement

The problem we solve is: How to build a proper 4-edge coloring of $K_{4,4}$ so that after removing the edges of 2 colors, removing two vertices $a \in A, b \in B$ and all corresponding edges, there is at least one perfect matching on the remaining graph. There are $4 * 4 = 16$ different combinations of a, b , we refer to them as *scenarios*.

Below we present two algorithm creating $K_{4,4}$ proper 4-edge coloring. Both take as input a perfect matching of $K_{3,3}$. Let us consider $K_{3,3} := (V_1, V_2, E)$ with $V_1 := \{1, 2, 3\}$ and $V_2 := \{4, 5, 6\}$ and $E := V_1 \times V_2$. Let M be a perfect matching on $K_{3,3}$, $N(M) := \{N_1, N_2, N_3\}$ the set of neighbors of M in the intersection graph, and (u_i, v_i) the common edge between M and N_i .

³Note: either $|u \cap v| = 1$ or $|u \cap v| = 0$

The intersection graph of two distinct 1-factorizations of $K_{3,3}$ begets $K_{3,3}$

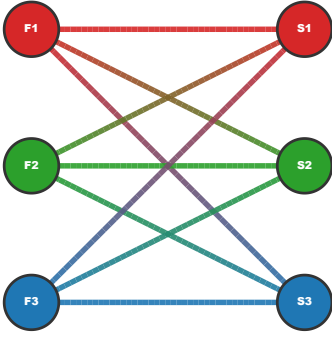


Figure 3: FLAT and STAR are two different - incompatible - proper edge coloring of $K_{3,3}$. Each perfect matching shares exactly one edge with all 3 perfect matching from the other proper edge coloring.

2.4 Problem reproduction

The **CROSS** algorithm produce unsuitable proper edge coloring by introducing systematically the observations of section 2.2.

- Inputs: M , a perfect matching on $K_{3,3}$
- Output: $C' := \{M'_1, M'_2, M'_3, M'_4\}$, a 1-factorization of $K_{4,4}$ (corresponds to a proper edge coloring).
- Step 1: $V'_1 := V_1 \cup \{a\}$, $V'_2 := V_2 \cup \{b\}$
- Step 2: let $M'_1 := M \cup \{(a, b)\}$
- Step 3: let $M'_{i+1} := N_i \setminus \{(u_i, v_i)\} \cup \{(a, v_i), (u_i, b)\}$
 $\forall i := 1, 2, 3$

The CROSS algorithm applied to all 6 perfect matching results systematically to 8 scenarios with two fair pairings available, and 8 scenarios with no fair pairings available, just like in figure 1.

2.5 Solution

The **ROT** algorithm avoids pointed observations by rotating the edges of a perfect matching. Only Step 3 is different. Let us index the two set $V'_1 = (a, 1, 2, 3) = (v'_{1,0}, \dots, v'_{1,3})$ and $V'_2 = (b, 6, 5, 4) = (v'_{2,0}, \dots, v'_{2,3})$. And then

- Step 3: For $i = 0, 1, 2, 3$,

$$M'_{i+1} = \{(v'_{1,j}, v'_{2,(j-i) \bmod 4}) : j = 0, 1, 2, 3\}$$

The Table 1 summarizes which of the 6 fair perfect matching from valve list is available for the last round with respect to all scenarios. The ROT algorithm indeed

ensures the existence of a fair matching for the last round. Appendices A.2 and A.3 provides visual of the removal scenarios for F1_CROSS and S1_CROSS. Appendix A.4 contains a table for **expected matching** with respect to all possible scenarios across every color ordering of each 12 outputs. It is build using cycle basis color analysis. See GitHub⁴ for more details. The original major appears to use F1_CROSS and the current major appears to also use F1_CROSS, but with color 0,3 played first. Interesting enough, it yields similar problematic scenarios.

3 Simulation & Evaluation

For evaluation we use RSTT[4], a python package for simulation of competitive system with high parameterization options. In the simulation, teams have a floating value representing *skill or level in the game*. The outcome of a game is computed by a *Solver* using the teams levels. For a tournament run, we can specify *seeding* as a ranking independant from the levels, and the *matching strategy to generate games* during the event. We perform a comparative study of tournament system with identical aim: qualify 8 teams mong 16.

- *4Groups4* typical sport tournament with round robin among 4 teams, best two of each groups qualifies.
- *2Groups8* variant used by the IIHF for ice hockey world championship, best 4 of both groups qualifies.
- *Original* system in Figure 1, and *Original-NTB*, a version without tiebreaker.
- *Major* current Counter-Strike major, and *Major-NTB* without tiebreaker.
- *Ema* Expected Matching Approach (ROT) with $M = S_1$
- *FixedDefault* the default RSTT implementation using Valve list for the last round pairings.

3.1 Parameters Matrix

Level models: GPR_teams (real Elo ratings, Riot 2024)⁵, Gaussian ($\mu = 1500, \sigma = 250$), Pareto ($\alpha = 5.5$), Weibull ($\alpha = 1.2, \beta = 7.0$).

Seeding classes: RANDOM, ACR2 - preserves top8, ACR3 preserves top4 and bottom4, ACR5 - preserves top5 and bottom5. Seed class quality is shown in Appendix B.1. We use 100 randomly sampled seeds from each classes.

Solver: BetterWin (deterministic, higher level wins), Logistic (stochastic, uses Elo formula for win

⁴https://github.com/Ematrion/SwissBracketMatching/blob/main/1_GraphColoring.ipynb

⁵<https://10esports.com/en-GB/gpr/2024/worlds?events=112966669920590211>

Table 1: last round expected perfect matching per covering and removal scenario. Value indicate Valve priority rank(s) of the available matching, dash indicates no fair pairing.

Coloring	(1,5)	(1,6)	(1,7)	(1,8)	(2,5)	(2,6)	(2,7)	(2,8)	(3,5)	(3,6)	(3,7)	(3,8)	(4,5)	(4,6)	(4,7)	(4,8)
F1_CROSS	5,6	5,6	—	—	5,6	5,6	—	—	—	—	4,6	4,6	—	—	4,6	4,6
F2_CROSS	2,5	—	1,3	—	—	3,5	—	1,6	—	3,5	—	1,6	3,4	—	5,6	—
F3_CROSS	—	1,6	1,6	—	4,6	—	—	1,3	—	3,4	3,4	—	3,5	—	—	2,4
S1_CROSS	5,6	5,6	—	—	5,6	5,6	—	—	—	—	4,6	4,6	—	—	4,6	4,6
S2_CROSS	1,6	—	2,4	—	—	4,6	—	2,5	3,4	—	5,6	—	—	3,5	—	1,6
S3_CROSS	—	2,5	2,5	—	3,5	—	—	2,4	3,5	—	—	2,4	—	3,4	3,4	—
F1_ROT	5	5	4	4	5	4	4	4	4	4	4	4	4	4	4	1
F2_ROT	4	3	3	1	4	3	6	1	4	6	6	1	4	6	6	5
F3_ROT	3	6	5	5	6	6	5	5	6	6	4	5	6	6	4	4
S1_ROT	5	5	6	6	4	5	6	6	4	4	6	6	4	4	6	2
S2_ROT	6	3	3	3	6	6	3	3	6	6	6	3	6	6	6	6
S3_ROT	3	4	5	2	3	4	4	2	6	4	4	2	6	4	4	3

probability[8]), MinEffort (Logistic variation where top-4 best teams have a 40% level penalty in the first two rounds).

We produce 20 samples: $20 \times 4 \times 4 \times 100 \times 3 = 96\,000$ tournaments per system; 768 000 total.

3.2 Metrics

We asses tournament ability to qualify the best teams, by comparing qualifications rate to a round robin of all 16 teams. Baseline was computed using 2000 tournaments run. Win probability matrix (Logistic solver) of each level models and Round-Robin qualification profiles are shown in Appendix B.1. After each tournament several metrics are computed to assess its outcome quality. Each metrics uses a ranking reference. We compute each metric twice, once using the *level ground-truth* and once using the *tournament seeds*.

Top-8 Precision, TOP8 (higher better ↑): fraction of qualified teams from the reference top-8.

Normalized Utility Loss, NUL (lower better ↓): For each Level model we computed for each teams the probabilty of top8 in Round Robin over 2000 simulation run. $NUL = (MaxUtility - TournamentUtility)/MaxUtility$ where *Max Utility* is the sum of the top8 probability of the 8 highest level teams and *Tournament Utility* is the sum of the top8 probability of the qualified teams (using the rank of the reference).

Fairness (↓): average rank of each game winner.[3]

Balance (↓): average absolute rank difference of each game opponents.[3]

Last-Round Matching Index: Valve’s priority rank (1,...,15) of the Round 5 pairing; random baseline 6/15 = 40% of fair matchings.

4 Results

Appendix C contains tables with all metrics with respect to different parameters combinations. In this section we

focus on the questions: does the Expected Matching Approach improve last round fairness and does it result in better competition quality?.

4.1 Last-Round Matching Quality

Table 2 shows Fairness rate in the last round.

Table 2: Round-5 Top-6 Preferred Pairing Rate (stochastic solvers).

System	Level (%)	Seed (%)
Original	41.5	35.4
Major	42.4	37.7
Ema	54.2	95.2
Original-NTB	53.8	91.0
Major-NTB	54.7	95.4
FixedDefault	54.4	96.0
<i>Random baseline</i>	<i>40.0</i>	<i>40.0</i>

EMA and NTB variants reach $\approx 54\%$ preferred-pairing rate (level), 14 points above the random baseline and 12 points above Original/Major ($\approx 42\%$). The seed-perspective gap is far more dramatic: EMA/NTB variants reach 91-96% while tiebreaker systems fall below the random baseline (35-38%).

4.2 Qualification Accuracy

Table 3 presents results for stochastic solvers, level perspective.

All Swiss variants reach 79-81% top-8 precision. The Current Major and Original variant are outperformed by the 2Groups8 baseline. Removing tiebreakers consistently improves precision regardless of the underlying matching algorithm. Note that EMA and NTB variants are *independent* approaches: EMA improves matching structure; NTB removes tiebreaker bias.

EMA achieves best overall Top8 and NUL, but also second best Fairness among Swiss Bracket systems.

Table 3: System performance – Stochastic Solvers, Level perspective. Top8 $\uparrow$, NUL $\downarrow$, Fairness $\downarrow$, Bal $\downarrow$.

System	Top8	NUL	Fair	Bal
4Groups4	0.758	0.151	5.335	6.234
2Groups8	0.797	0.112	5.514	5.837
Original	0.790	0.117	5.861	5.243
Major	0.788	0.120	5.877	5.239
Ema	0.806	0.105	5.807	5.382
Original-NTB	0.806	0.105	5.806	5.382
Major-NTB	0.804	0.106	5.819	5.390
FixedDefault	0.803	0.108	5.821	5.339

5 Discussion & Conclusion

From the simulation we can confirm that Swiss Bracket is a tournament that better qualify teams based on their level. The Original and current Major are the worst designs, yet out perform notably groups stage of 4 teams, classic sport tournament. The 2023 speed-up did not improve the quality. The Expected Matching approach does improve significantly the fairness of the tournament, but so does the *NTB* variants. Instead of one fair last round every 3 tournament editions, EMA sees one a priori unfair last round every 20 tournament!

The expected matching approach addresses pairing only in the 0-0 and 1-1 rounds, for a total of 8 games out of 33. Additionally it provides minimum alternatives for pairing. It provides 4 precomputed pairings assuming it is enough for the 1-1 round. However, observation (3) is a general remark that can be used as a heuristic to validate the output of general matching algorithms at any stage of the tournament.

5.1 Future Work

- **Generalization** SwissBracket can be played for $n=16, 32, \dots$ 32 is not as challenging since the last round involves 10 teams that have previously faced each other 3 times.
- **Coloring Class** There seems to be more proper edge coloring of $K_{4,4}$ than the one produced by the CROSS and ROT algorithms. Which ones and how do they suit in the format.
- **Formal Problem** Let us consider positives integers $n_0 > n_1 > \dots > n_N$ and their associated complete bipartite graphs $K_{n_i, n_i} \supset K_{n_j, n_j} \forall i > j$. For which integer $d_0, d_1, \dots, d_k$, such that $\sum_{l=1}^k d_l = n_N$ is it possible to build perfect matching $M_{n_m, d_0}, \dots, M_{n_m, d_m}$ on K_{n_m, n_m} so that $\bigcap M_{n_i, d_j} = \emptyset$.⁶
- Explore color partition of cycle basis for classification of proper edge coloring.

⁶EMA addresses the case $N = 1, n_0 = 4, n_1 = 3, d_0 = 2, d_1 = 1$

- Potential tiebreaking system to improve EMA.

5.2 Disclaimer

During this work, we naturally rediscover, by solving a practical problem, the role of 2-color cycles in a proper edge coloring. This is not a novelty; the notion of *acyclic edge coloring*[7] exists in the literature. This research did not rely on any previous literature, except base concept of graph theory and we had no previous knowledge on such related work.

5.3 Conclusion

Even with minimal intervention and further optimisation, EMA performs very well according to all metrics. But a more important take away is the quality of the format, significantly above all alternatives. Swiss Bracket should be adopted in any competition involving a main stage event before play-offs.

References

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A Graph Colorings

A.1 Proper 4-edge Colorings of $K_{4,4}$

Figure 4 shows all twelve proper 4-edge colorings of $K_{4,4}$ produced by the two algorithms. CROSS refer to an output by the CROSS algorithm, ROT to an output by the rotation algorithm. The first two character (for example F1) refers to the input perfect matching.

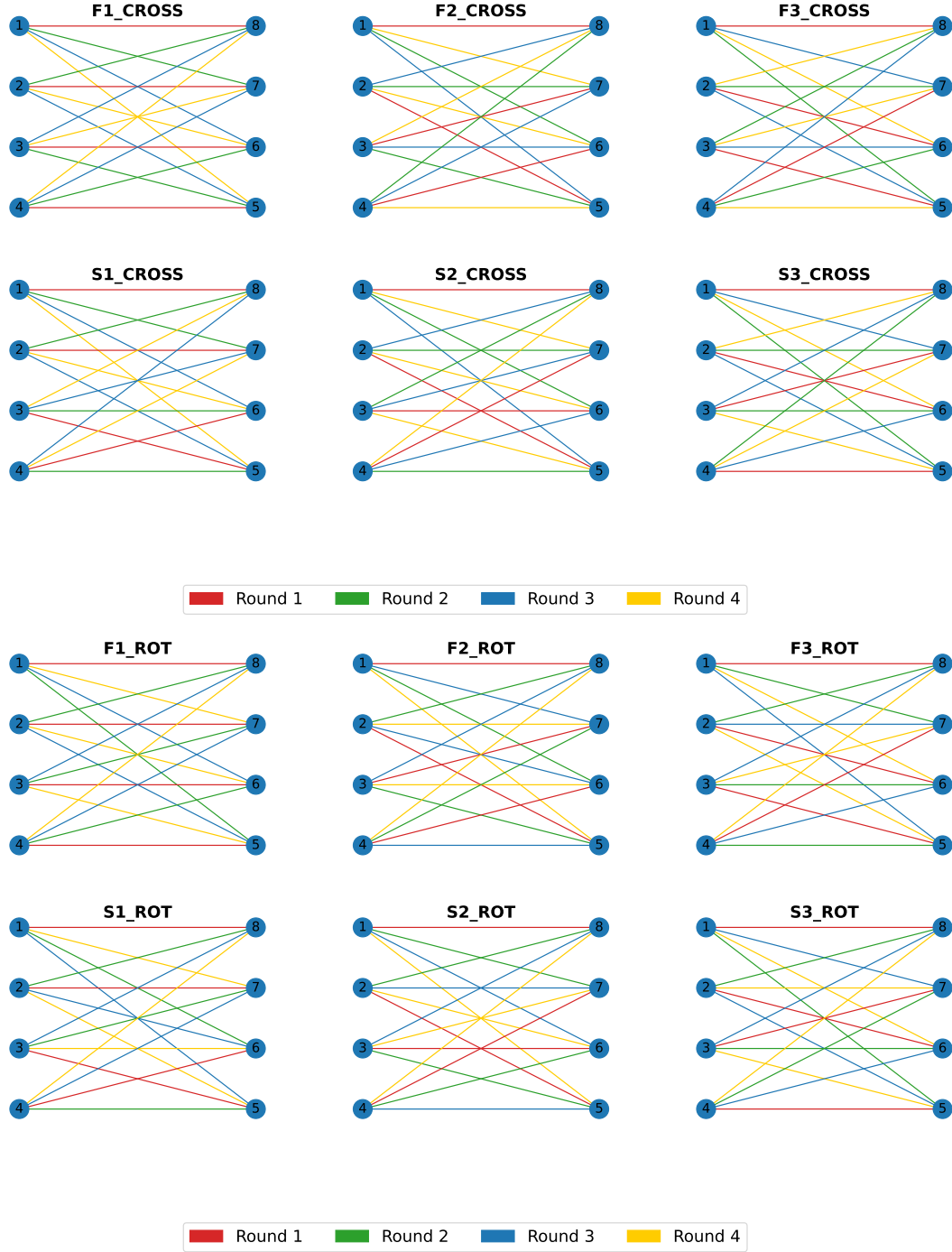


Figure 4: Top: the six $K_{4,4}$ colorings produced by the CROSS algorithm (F1_CROSS through S3_CROSS). Bottom: the six colorings produced by the ROT algorithm (F1_ROT through S3_ROT).

A.2 CROSS Algorithm: Representative Scenario Analysis

Figure 5 shows the Round-5 residual graph for each of the 16 removal scenarios under coloring F1_CROSS. The first two round edges are removed; then one team from each side ($a \in A$, $b \in B$) is eliminated. Each panel label gives the removed pair. Scenarios (1,7), (1,8), (2,7), (2,8), (3,5), (3,6), (4,5), (4,6) yield no preferred matching, consistent with Table ?? . The viral scenario is the (3,6) case, with indeed no perfect matching.

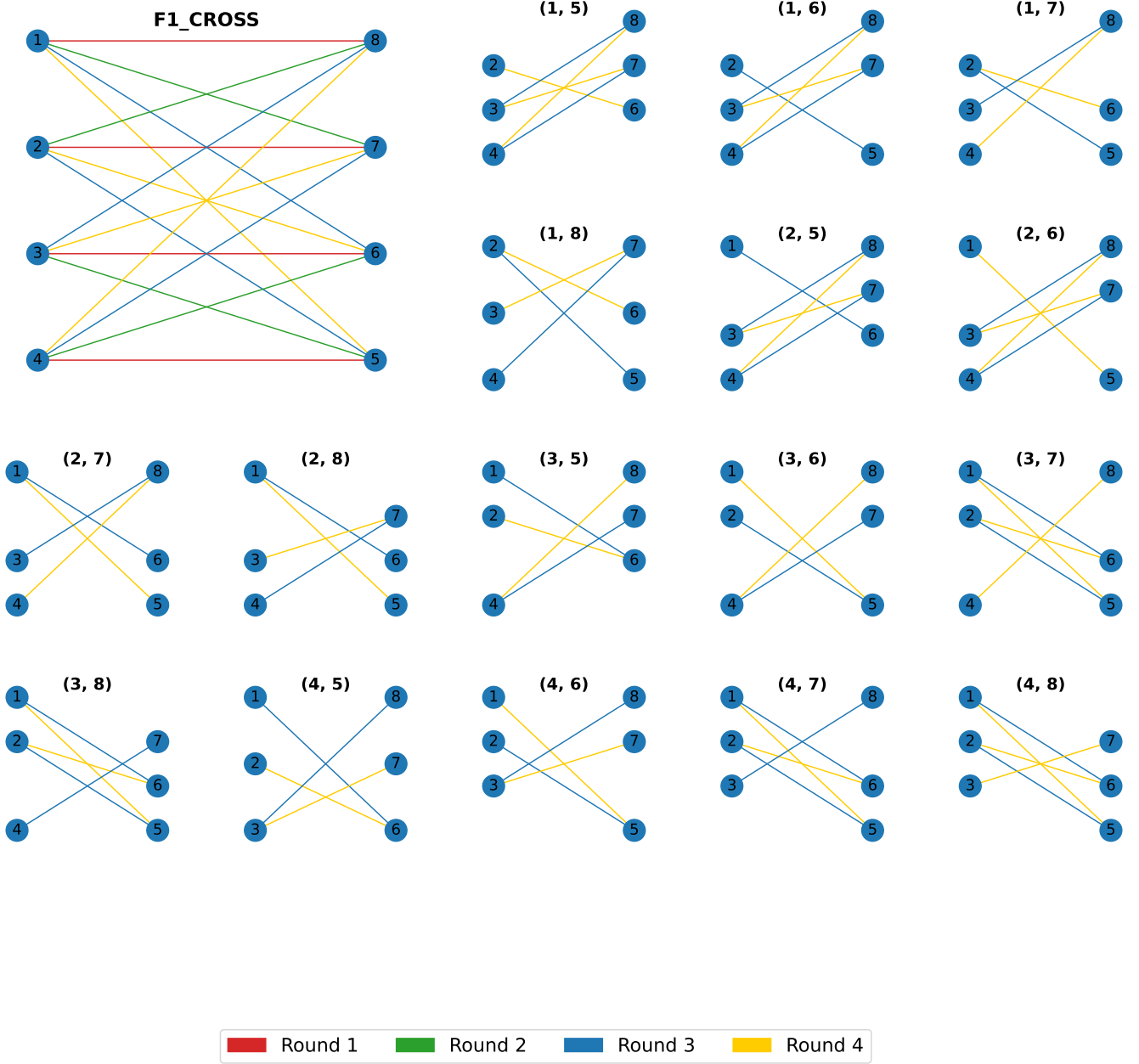


Figure 5: F1_CROSS - Round-5 residual graphs for all 16 removal scenarios. Each panel label gives the removed pair (a, b) , $a \in \{1, 2, 3, 4\}$, $b \in \{5, 6, 7, 8\}$. Empty graphs indicate no preferred matching exists.

A.3 Rotation Algorithm: Representative Scenario Analysis

Figure 6 shows the same analysis for coloring F1_ROT. Every scenario yields exactly one perfect matching, confirming the structural guarantee of the rotation algorithm.

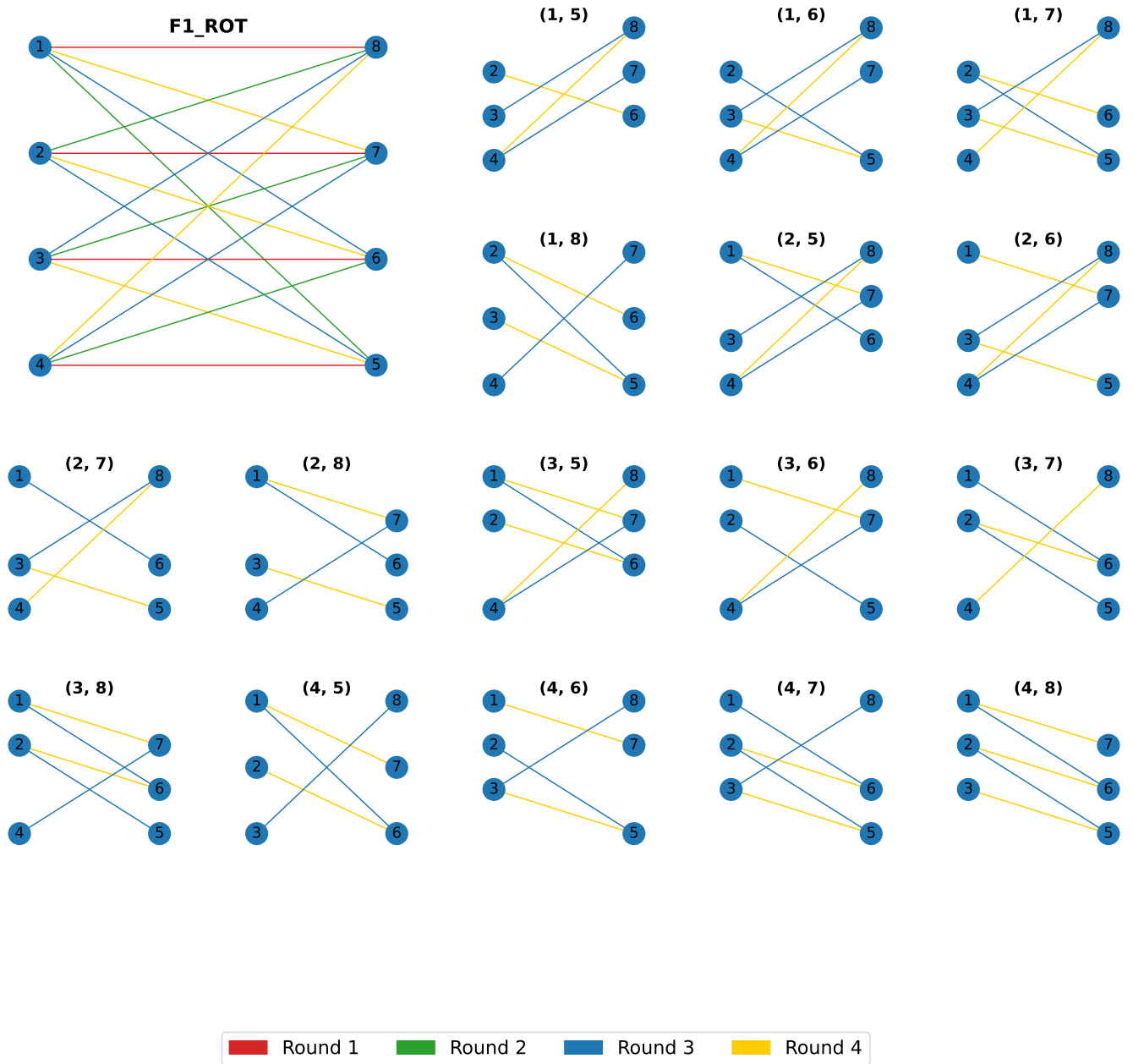


Figure 6: F1_ROT - Round-5 residual graphs for all 16 removal scenarios.

A.4 Round-Ordering Analysis

Tables 4 and 5 give the complete round-ordering analysis. For each coloring and each choice of which two of the four perfect matchings are scheduled first (*first two PMs*, zero-indexed), the table lists the available preferred matching for each of the 16 removal scenarios. A dash denotes no preferred matching; two values indicate two options.

Table 4: Round-ordering, ROT analysis. Columns are removal scenarios (a, b) .

Coloring	First 2 PM	1,5	1,6	1,7	1,8	2,5	2,6	2,7	2,8	3,5	3,6	3,7	3,8	4,5	4,6	4,7	4,8
F1_ROT	(0,1)	5	5	4	4	5	4	4	4	4	4	4	4	4	4	4	1
F1_ROT	(0,2)	1,6	—	2,4	—	—	4,6	—	2,5	3,4	—	5,6	—	—	3,5	—	1,6
F1_ROT	(0,3)	1	5	5	5	5	5	5	5	5	5	5	4	5	5	4	4
F1_ROT	(1,2)	4	4	4	1	4	4	1	1	4	1	1	1	1	1	1	1
F1_ROT	(1,3)	—	3,5	—	1,6	2,5	—	1,3	—	—	1,2	—	3,4	1,6	—	2,4	—
F1_ROT	(2,3)	1	1	1	1	1	1	1	5	1	1	5	5	1	5	5	5
F2_ROT	(0,1)	4	3	3	1	4	3	6	1	4	6	6	1	4	6	6	5
F2_ROT	(0,2)	2,4	2,4	—	—	—	—	4,6	4,6	5,6	5,6	—	—	—	—	3,5	3,5
F2_ROT	(0,3)	5	2	3	4	5	3	3	4	5	3	3	1	4	3	3	1
F2_ROT	(1,2)	4	6	6	5	1	6	6	5	1	6	2	5	1	2	2	5
F2_ROT	(1,3)	—	—	3,5	3,5	1,3	1,3	—	—	—	—	1,2	1,2	2,4	2,4	—	—
F2_ROT	(2,3)	1	2	2	5	1	2	2	4	5	2	2	4	5	2	3	4
F3_ROT	(0,1)	3	6	5	5	6	6	5	5	6	6	4	5	6	6	4	4
F3_ROT	(0,2)	—	1,6	1,6	—	4,6	—	—	1,3	—	3,4	3,4	—	3,5	—	—	2,4
F3_ROT	(0,3)	3	3	1	1	3	3	5	1	3	3	5	5	3	6	5	5
F3_ROT	(1,2)	6	6	4	4	6	2	4	4	2	2	4	4	2	2	1	4
F3_ROT	(1,3)	3,5	—	—	2,4	—	2,5	2,5	—	1,2	—	—	5,6	—	1,6	1,6	—
F3_ROT	(2,3)	2	2	1	4	2	2	1	1	2	3	1	1	3	3	1	1
S1_ROT	(0,1)	5	5	6	6	4	5	6	6	4	4	6	6	4	4	6	2
S1_ROT	(0,2)	—	1,6	1,6	—	4,6	—	—	1,3	—	3,4	3,4	—	3,5	—	—	2,4
S1_ROT	(0,3)	5	1	3	3	5	5	3	3	5	5	3	6	5	5	6	6
S1_ROT	(1,2)	4	4	6	2	4	4	2	2	1	4	2	2	1	1	2	2
S1_ROT	(1,3)	3,5	—	—	2,4	—	2,5	2,5	—	1,2	—	—	5,6	—	1,6	1,6	—
S1_ROT	(2,3)	1	1	2	2	1	1	2	3	1	1	3	3	5	1	3	3
S2_ROT	(0,1)	6	3	3	3	6	6	3	3	6	6	6	3	6	6	6	6
S2_ROT	(0,2)	1,6	—	2,4	—	—	4,6	—	2,5	3,4	—	5,6	—	—	3,5	—	1,6
S2_ROT	(0,3)	3	3	2	2	3	3	3	2	3	3	3	3	6	3	3	3
S2_ROT	(1,2)	6	6	6	6	2	6	6	6	2	2	6	6	2	2	2	6
S2_ROT	(1,3)	—	3,5	—	1,6	2,5	—	1,3	—	—	1,2	—	3,4	1,6	—	2,4	—
S2_ROT	(2,3)	2	2	2	6	2	2	2	2	3	2	2	2	3	3	2	2
S3_ROT	(0,1)	3	4	5	2	3	4	4	2	6	4	4	2	6	4	4	3
S3_ROT	(0,2)	2,4	2,4	—	—	—	—	4,6	4,6	5,6	5,6	—	—	—	—	3,5	3,5
S3_ROT	(0,3)	2	5	5	6	3	5	5	6	3	5	5	2	3	4	5	2
S3_ROT	(1,2)	6	4	4	3	6	1	4	3	6	1	1	3	2	1	1	3
S3_ROT	(1,3)	—	—	3,5	3,5	1,3	1,3	—	—	—	—	1,2	1,2	2,4	2,4	—	—
S3_ROT	(2,3)	2	1	1	3	2	1	1	6	2	5	1	6	2	5	5	6

Table 5: Round-ordering, CROSS analysis. Columns are removal scenarios (a, b) .

Coloring	First	2	PM	1,5	1,6	1,7	1,8	2,5	2,6	2,7	2,8	3,5	3,6	3,7	3,8	4,5	4,6	4,7	4,8
F1_CROSS	(0,1)			5,6	5,6	—	—	5,6	5,6	—	—	—	—	4,6	4,6	—	—	4,6	4,6
F1_CROSS	(0,2)			1,6	—	2,4	—	—	4,6	—	2,5	3,4	—	5,6	—	—	3,5	—	1,6
F1_CROSS	(0,3)			—	2,5	2,5	—	3,5	—	—	2,4	3,5	—	—	2,4	—	3,4	3,4	—
F1_CROSS	(1,2)			4,6	—	—	1,3	—	1,6	1,6	—	—	1,6	1,6	—	1,2	—	—	5,6
F1_CROSS	(1,3)			—	3,5	—	1,6	2,5	—	1,3	—	—	1,2	—	3,4	1,6	—	2,4	—
F1_CROSS	(2,3)			—	—	1,2	1,2	—	—	1,2	1,2	1,3	1,3	—	—	1,3	1,3	—	—
F2_CROSS	(0,1)			2,5	—	1,3	—	—	3,5	—	1,6	—	3,5	—	1,6	3,4	—	5,6	—
F2_CROSS	(0,2)			—	3,4	3,4	—	—	3,4	3,4	—	4,6	—	—	1,3	4,6	—	—	1,3
F2_CROSS	(0,3)			2,4	2,4	—	—	—	—	4,6	4,6	5,6	5,6	—	—	—	—	3,5	3,5
F2_CROSS	(1,2)			—	—	3,5	3,5	1,3	1,3	—	—	—	—	1,2	1,2	2,4	2,4	—	—
F2_CROSS	(1,3)			1,2	—	—	5,6	1,2	—	—	5,6	—	2,5	2,5	—	—	2,5	2,5	—
F2_CROSS	(2,3)			—	4,6	—	2,5	1,6	—	2,4	—	1,6	—	2,4	—	—	1,2	—	3,4
F3_CROSS	(0,1)			—	1,6	1,6	—	4,6	—	—	1,3	—	3,4	3,4	—	3,5	—	—	2,4
F3_CROSS	(0,2)			1,3	1,3	—	—	—	—	3,5	3,5	—	—	3,5	3,5	5,6	5,6	—	—
F3_CROSS	(0,3)			3,4	—	5,6	—	3,4	—	5,6	—	—	4,6	—	2,5	—	4,6	—	2,5
F3_CROSS	(1,2)			—	1,2	—	3,4	—	1,2	—	3,4	2,5	—	1,3	—	2,5	—	1,3	—
F3_CROSS	(1,3)			—	—	4,6	4,6	2,4	2,4	—	—	2,4	2,4	—	—	—	—	1,2	1,2
F3_CROSS	(2,3)			3,5	—	—	2,4	—	2,5	2,5	—	1,2	—	—	5,6	—	1,6	1,6	—
S1_CROSS	(0,1)			5,6	5,6	—	—	5,6	5,6	—	—	—	—	4,6	4,6	—	—	4,6	4,6
S1_CROSS	(0,2)			2,5	—	1,3	—	—	3,5	—	1,6	—	3,5	—	1,6	3,4	—	5,6	—
S1_CROSS	(0,3)			—	1,6	1,6	—	4,6	—	—	1,3	—	3,4	3,4	—	3,5	—	—	2,4
S1_CROSS	(1,2)			3,5	—	—	2,4	—	2,5	2,5	—	1,2	—	—	5,6	—	1,6	1,6	—
S1_CROSS	(1,3)			—	4,6	—	2,5	1,6	—	2,4	—	1,6	—	2,4	—	—	1,2	—	3,4
S1_CROSS	(2,3)			—	—	1,2	1,2	—	—	1,2	1,2	1,3	1,3	—	—	1,3	1,3	—	—
S2_CROSS	(0,1)			1,6	—	2,4	—	—	4,6	—	2,5	3,4	—	5,6	—	—	3,5	—	1,6
S2_CROSS	(0,2)			—	3,4	3,4	—	—	3,4	3,4	—	4,6	—	—	1,3	4,6	—	—	1,3
S2_CROSS	(0,3)			1,3	1,3	—	—	—	—	3,5	3,5	—	—	3,5	3,5	5,6	5,6	—	—
S2_CROSS	(1,2)			—	—	4,6	4,6	2,4	2,4	—	—	2,4	2,4	—	—	—	—	1,2	1,2
S2_CROSS	(1,3)			1,2	—	—	5,6	1,2	—	—	5,6	—	2,5	2,5	—	—	2,5	2,5	—
S2_CROSS	(2,3)			—	3,5	—	1,6	2,5	—	1,3	—	—	1,2	—	3,4	1,6	—	2,4	—
S3_CROSS	(0,1)			—	2,5	2,5	—	3,5	—	—	2,4	3,5	—	—	2,4	—	3,4	3,4	—
S3_CROSS	(0,2)			2,4	2,4	—	—	—	—	4,6	4,6	5,6	5,6	—	—	—	—	3,5	3,5
S3_CROSS	(0,3)			3,4	—	5,6	—	3,4	—	5,6	—	—	4,6	—	2,5	—	4,6	—	2,5
S3_CROSS	(1,2)			—	1,2	—	3,4	—	1,2	—	3,4	2,5	—	1,3	—	2,5	—	1,3	—
S3_CROSS	(1,3)			—	—	3,5	3,5	1,3	1,3	—	—	—	—	1,2	1,2	2,4	2,4	—	—
S3_CROSS	(2,3)			4,6	—	—	1,3	—	1,6	1,6	—	—	1,6	1,6	—	1,2	—	—	5,6

B Simulation Parameters

B.1 Seed Classes

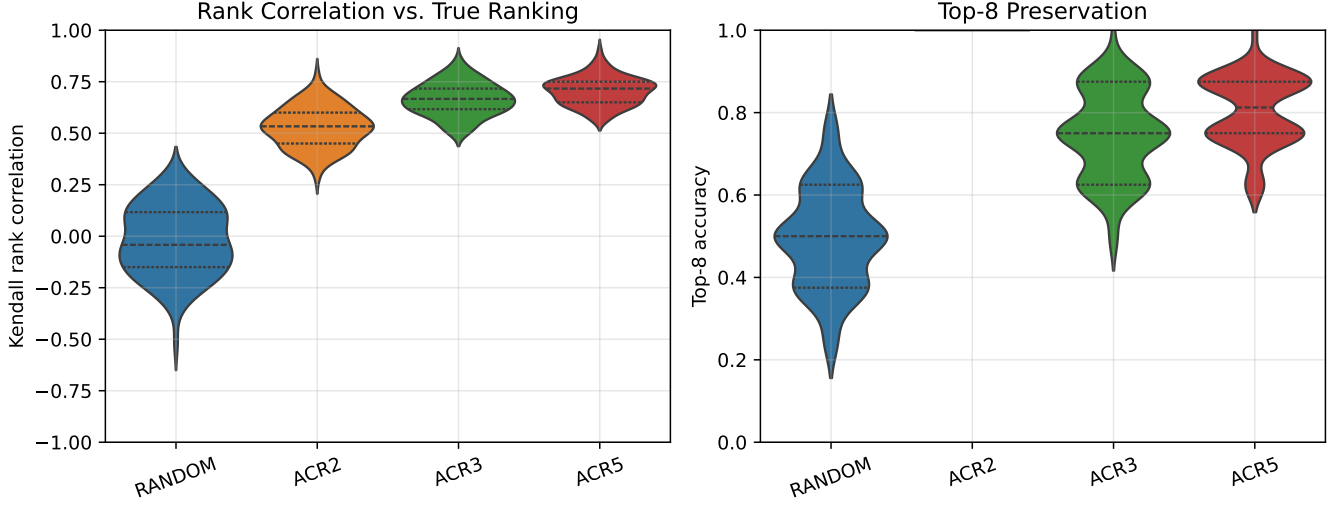


Figure 7: Seeding quality distribution across 100 variants per class. Left: Kendall rank correlation. Right: top-8 accuracy. RANDOM: complete shuffle; ACR2: top-8/bottom-8 preservation; ACR3: preserves three segments (top-4, middle-8, and bottom-4); ACR5 preserves top5; middle-6 and bottom-5. ACR2 has a top-8 preservation of 1.0 by construction.

B.2 Level Model & Qualification Baselines



Figure 8: Upper row: direct confrontation win probability matrix, teams ordered by level rank. Lower row: top-8 qualification probability by rank under round robin baseline (2000 simulations using Logistic solver). GPR_teams uses real Elo ratings (Riot 2024 Global Power Ranking), Gaussian: $\mu = 1500, \sigma = 250$, Pareto: $\alpha = 5.5$, Weibull: $\alpha = 1.2, \beta = 7.0$

C Simulation Results

All values are means over $20 \times 3 \times 4 \times 4 \times 100 = 96\,000$ tournaments per system. **NUL**: Normalized Utility Loss ($\downarrow$). **Top8**: top-8 precision ($\uparrow$). **Fair**: average winner rank ($\downarrow$). **Bal**: average rank difference ($\downarrow$). **LR**: metric restricted to Round 5. Each table pair shows the **level** perspective (left) and **seed** perspective (right).

Table A1-A2: BetterWin Solver

<i>Level perspective</i>						<i>Seed perspective</i>					
System	NUL	Top8	Fair	Bal	LR Fair	System	NUL	Top8	Fair	Bal	LR Fair
4Groups4	0.034	0.912	4.383	6.234	—	4Groups4	0.169	0.751	5.334	6.667	—
2Groups8	0.014	0.947	4.581	5.837	—	2Groups8	0.167	0.753	5.632	6.000	—
Original	0.011	0.956	5.326	4.344	5.935	Original	0.164	0.758	6.105	5.518	6.508
Major	0.011	0.955	5.332	4.338	5.946	Major	0.161	0.764	6.075	5.376	6.618
Ema	0.014	0.945	5.335	4.330	5.985	Ema	0.164	0.757	6.085	5.910	6.672
Orig-NTB	0.014	0.946	5.335	4.330	5.978	Orig-NTB	0.161	0.761	6.083	5.855	6.611
Maj-NTB	0.016	0.941	5.341	4.318	6.007	Maj-NTB	0.167	0.751	6.093	5.704	6.758
FixedDef	0.015	0.943	5.339	4.317	5.923	FixedDef	0.164	0.758	6.095	5.772	6.524

Table A3-A4: Stochastic Solvers

<i>Level perspective</i>						<i>Seed perspective</i>					
System	NUL	Top8	Fair	Bal	LR Fair	System	NUL	Top8	Fair	Bal	LR Fair
4Groups4	0.151	0.758	5.335	6.234	—	4Groups4	0.241	0.679	6.029	6.667	—
2Groups8	0.112	0.797	5.514	5.837	—	2Groups8	0.226	0.693	6.234	6.000	—
Original	0.117	0.790	5.861	5.243	6.438	Original	0.227	0.691	6.410	5.916	6.847
Major	0.120	0.788	5.877	5.239	6.426	Major	0.226	0.696	6.419	5.937	6.808
Ema	0.105	0.806	5.807	5.382	6.422	Ema	0.203	0.720	6.308	6.455	6.751
Orig-NTB	0.105	0.806	5.806	5.382	6.438	Orig-NTB	0.202	0.719	6.307	6.445	6.763
Maj-NTB	0.106	0.804	5.819	5.390	6.406	Maj-NTB	0.200	0.724	6.307	6.461	6.719
FixedDef	0.108	0.803	5.821	5.339	6.378	FixedDef	0.208	0.713	6.332	6.303	6.660

Tables A5-A6: Top-8 Precision by Seeding Class

<i>Level perspective</i>						<i>Seed perspective</i>					
System	RND	ACR2	ACR3	ACR5	Mean	System	RND	ACR2	ACR3	ACR5	Mean
4Groups4	0.719	0.792	0.752	0.768	0.758	4Groups4	0.491	0.792	0.693	0.739	0.679
2Groups8	0.784	0.811	0.794	0.798	0.797	2Groups8	0.490	0.811	0.713	0.756	0.693
Original	0.776	0.800	0.791	0.794	0.790	Original	0.494	0.800	0.714	0.756	0.691
Major	0.777	0.802	0.783	0.789	0.788	Major	0.493	0.802	0.727	0.762	0.696
Ema	0.774	0.841	0.796	0.811	0.806	Ema	0.491	0.841	0.753	0.794	0.720
Orig-NTB	0.774	0.841	0.796	0.812	0.806	Orig-NTB	0.492	0.841	0.753	0.792	0.719
Maj-NTB	0.773	0.842	0.792	0.810	0.804	Maj-NTB	0.493	0.842	0.764	0.798	0.724
FixedDef	0.776	0.830	0.796	0.809	0.803	FixedDef	0.493	0.830	0.744	0.786	0.713

Tables A7-A8: NUL by Seeding Class

<i>Level perspective</i>						<i>Seed perspective</i>					
System	RND	ACR2	ACR3	ACR5	Mean	System	RND	ACR2	ACR3	ACR5	Mean
4Groups4	0.188	0.132	0.150	0.135	0.151	4Groups4	0.437	0.174	0.197	0.157	0.241
2Groups8	0.122	0.106	0.111	0.108	0.112	2Groups8	0.436	0.157	0.173	0.137	0.226
Original	0.130	0.114	0.114	0.112	0.117	Original	0.432	0.164	0.173	0.138	0.227
Major	0.129	0.112	0.122	0.116	0.120	Major	0.434	0.164	0.168	0.138	0.226
Ema	0.131	0.088	0.107	0.094	0.105	Ema	0.436	0.124	0.143	0.108	0.203
Orig-NTB	0.131	0.089	0.106	0.094	0.105	Orig-NTB	0.435	0.125	0.142	0.108	0.202
Maj-NTB	0.131	0.087	0.110	0.095	0.106	Maj-NTB	0.433	0.124	0.137	0.107	0.200
FixedDef	0.130	0.095	0.108	0.097	0.108	FixedDef	0.434	0.135	0.149	0.113	0.208

Tables A9-A10: Top-8 Precision by Population Model

<i>Level perspective</i>						<i>Seed perspective</i>					
System	GPR	Gauss	Pareto	Weibull	Mean	System	GPR	Gauss	Pareto	Weibull	Mean
4Groups4	0.701	0.790	0.738	0.802	0.758	4Groups4	0.641	0.701	0.671	0.702	0.679
2Groups8	0.746	0.824	0.772	0.845	0.797	2Groups8	0.661	0.711	0.684	0.714	0.693
Original	0.738	0.817	0.768	0.838	0.790	Original	0.661	0.708	0.685	0.710	0.691
Major	0.734	0.816	0.766	0.836	0.788	Major	0.665	0.713	0.689	0.716	0.696
Ema	0.759	0.829	0.785	0.849	0.806	Ema	0.691	0.736	0.715	0.738	0.720
Orig-NTB	0.759	0.828	0.785	0.849	0.806	Orig-NTB	0.690	0.736	0.713	0.738	0.719
Maj-NTB	0.759	0.827	0.783	0.849	0.804	Maj-NTB	0.697	0.737	0.720	0.742	0.724
FixedDef	0.757	0.826	0.781	0.847	0.803	FixedDef	0.685	0.730	0.707	0.731	0.713

Tables A11-A12: NUL by Population Model

<i>Level perspective</i>						<i>Seed perspective</i>					
System	GPR	Gauss	Pareto	Weibull	Mean	System	GPR	Gauss	Pareto	Weibull	Mean
4Groups4	0.194	0.131	0.141	0.140	0.151	4Groups4	0.263	0.229	0.228	0.245	0.241
2Groups8	0.149	0.097	0.105	0.097	0.112	2Groups8	0.241	0.217	0.213	0.232	0.226
Original	0.156	0.103	0.108	0.103	0.117	Original	0.241	0.220	0.212	0.235	0.227
Major	0.159	0.104	0.111	0.105	0.120	Major	0.243	0.217	0.213	0.232	0.226
Ema	0.138	0.094	0.095	0.093	0.105	Ema	0.215	0.197	0.188	0.210	0.203
Orig-NTB	0.138	0.094	0.095	0.093	0.105	Orig-NTB	0.215	0.196	0.188	0.210	0.202
Maj-NTB	0.138	0.094	0.097	0.094	0.106	Maj-NTB	0.212	0.195	0.186	0.207	0.200
FixedDef	0.140	0.096	0.099	0.095	0.108	FixedDef	0.220	0.202	0.194	0.216	0.208

Tables A13-A14: Round-5 Preferred Pairing Rate by Seeding Class (Stochastic Solvers, %)

<i>Level perspective</i>						<i>Seed perspective</i>					
System	RND	ACR2	ACR3	ACR5	Mean	System	RND	ACR2	ACR3	ACR5	Mean
Original	42.6	38.9	42.8	41.6	41.5	Original	33.3	37.9	34.1	36.1	35.4
Major	42.8	39.9	43.2	43.6	42.4	Major	34.1	39.5	38.9	38.3	37.7
Ema	41.2	77.9	45.2	52.5	54.2	Ema	93.0	95.2	96.3	96.3	95.2
Orig-NTB	41.6	76.8	45.0	51.8	53.8	Orig-NTB	91.3	93.3	90.3	89.1	91.0
Maj-NTB	41.9	76.6	46.8	53.5	54.7	Maj-NTB	94.6	93.9	96.8	96.2	95.4
FixedDef	41.5	75.6	45.8	54.8	54.4	FixedDef	93.9	96.1	97.2	97.0	96.0

Tables A15-A16: Round-5 Preferred Pairing Rate by Population Model (Stochastic Solvers, %)

<i>Level perspective</i>						<i>Seed perspective</i>					
System	GPR	Gauss	Pareto	Weibull	Mean	System	GPR	Gauss	Pareto	Weibull	Mean
Original	38.4	42.7	40.2	44.7	41.5	Original	34.5	35.1	35.1	36.7	35.4
Major	39.6	44.4	41.2	44.5	42.4	Major	35.8	39.0	36.7	39.3	37.7
Ema	54.5	53.1	53.8	55.4	54.2	Ema	94.4	95.8	95.0	95.7	95.2
Orig-NTB	53.5	53.6	53.2	54.8	53.8	Orig-NTB	92.4	89.0	91.4	91.1	91.0
Maj-NTB	56.6	52.8	53.5	56.0	54.7	Maj-NTB	94.4	96.1	94.8	96.2	95.4
FixedDef	55.2	53.3	53.3	55.8	54.4	FixedDef	95.3	96.8	95.5	96.5	96.0